

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Concentrated reliance flags a conformal vulnerability

RESEARCH QUESTION

Can validation-time feature reliance help identify gradient-boosted classifiers whose prediction-set coverage will deteriorate under shift?

16	0.853	77.1 pp
multiclass tasks in pooled endpoint	Spearman association with coverage loss	SALT payment-terms coverage loss

Problem

Conformal prediction targets inclusion of the true label in a prediction set. Its standard guarantee relies on exchangeability. Eight supply-chain tasks under a shared COVID-era temporal split show sharply different coverage losses.

Observed evidence

Across 16 multiclass tasks from 9 domains, concentration and coverage loss have Spearman $\rho = 0.853$ ($p < 0.001$; bootstrap 95% interval [0.50, 0.96]). This pool includes 8 original SALT tasks and 8 external tasks, including 4 null-shift controls.

Counterexamples matter

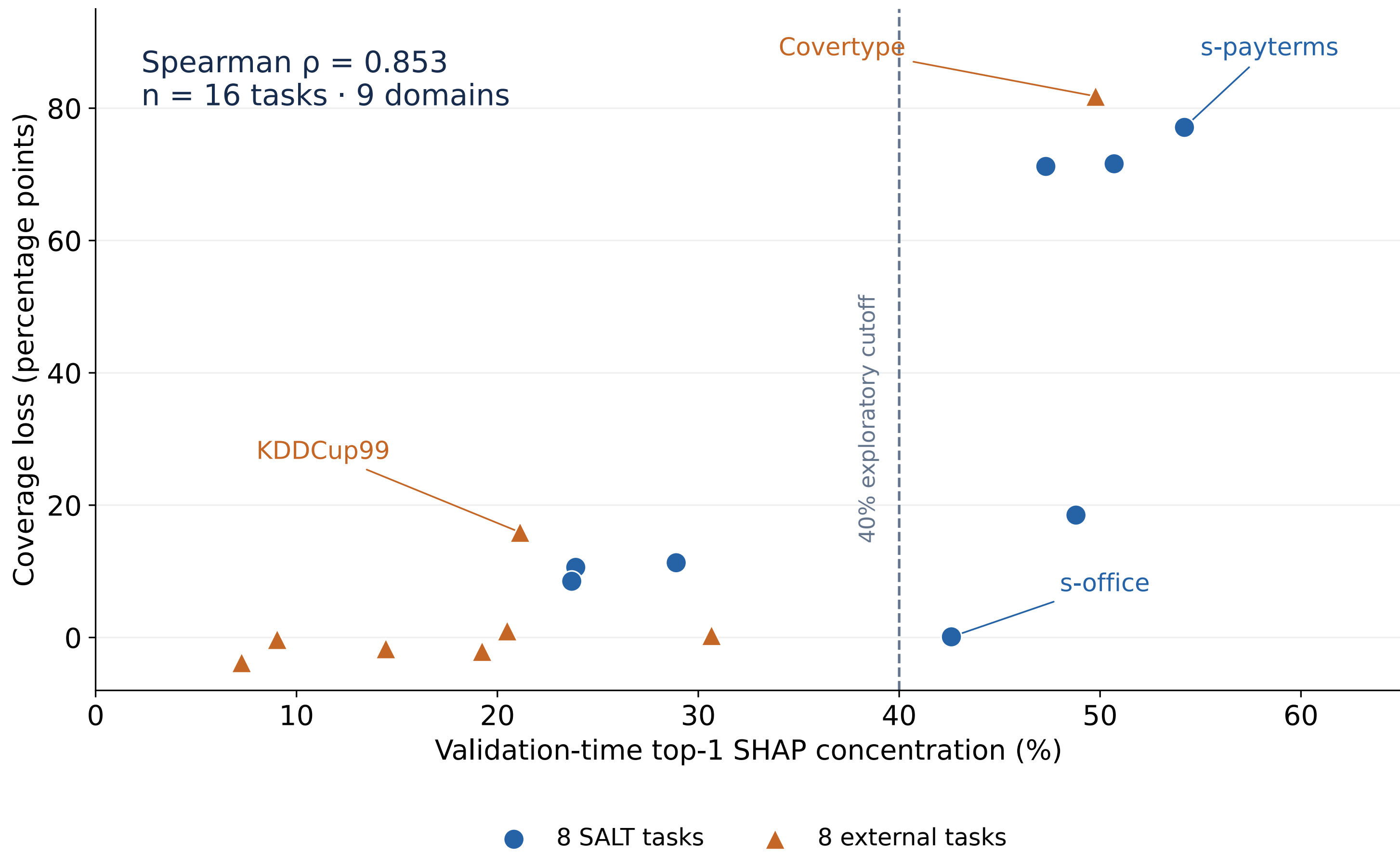
At the exploratory 40% cutoff, sales-office is a false positive ($C = 42.6\%$, loss 0.1 pp), while KDDCup99 is a seed-sensitive false negative ($C = 21.1\%$, mean loss 15.9 pp). Low concentration is not universal protection.

Diagnostic

Top-1 concentration C is the largest mean absolute SHAP importance divided by total importance. It is computed on validation data before deployment. High C indicates concentrated measured reliance, not the size or direction of a future shift.

Mechanism and boundary

An idealized additive probability model links damage to a dominant feature with worsening APS score and coverage bounds. Measured TreeSHAP concentration is a proxy; the theorem does not guarantee its deployment behavior.



Exact source data replotted: 8 SALT and 8 external multiclass tasks. The 40% cutoff is exploratory.

TAKEAWAY

Inspect which features a model relies on before trusting its future coverage. Use concentration to guide investigation and monitoring; confirm any threshold prospectively.

INTERPRETATION & LIMITS

Evidence is strongest for gradient-boosted multiclass APS models. The 40% cutoff is exploratory; the protective-feature check comes from one case. Pooled tasks include development tasks and null controls. MLP failures can occur at low concentration. Association does not identify a causal effect.